MILITARY SPECIFICATION

INTEGRATED ELECTRONIC CENTRALS AN/ASQ-19, AN/ASQ-19A, AN/ASQ-19B, and AN/ASQ-107

This amendment forms a part of Military Specification MIL-I-18309C (WEP) dated 1 May 1963 and has been approved by the Naval Air Systems Command, Department of the Navy.

By this amendment basic MIL-I-18309C(WEP) is changed to MIL-I-18309C(AS) and throughout the specification, Bureau of Naval Weapons shall be changed to Naval Air Systems Command.

* *Page 1*, Specification Title: Delete and substitute the title of this amendment.

* *Page 1*, paragraph 1.2: Delete and substitute

"**1.2 Classification.** The Integrated Electronic Centrals shall be of Four types which together shall include the following equipments:

Item	*Name*	*Type Designation*	*MTBF*	*MIL-STD-781 Test Plan*
1	Receiver-Transmitter, Radio	RT-546/ASQ-19	170	III
2	Receiver-Tarnsmitter, Radio	RT-546A/ASQ-19	170	III
3	Receiver-Transmitter, Radio	RT-546B/ASQ-19	170	III
4	Receiver-Transmitter, Radio	RT-853/ASQ-19	170	III
5	Receiver Transmitter, Radio	RT-793/ASQ	200	III
6	Receiver-Transmitter, Radio	RT-793A/ASQ	200	III
7	Receiver-Transmitter, Radio Decoder, Pulse	RT-547/ASQ-19 KY-312/ASQ-19	150	III
8	Coder-Receiver-Transmitter	KY-532/ASQ	450	IV

FSC 5895

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Item	*Name*	*Type Designation*	*MTBF*	*MIL–STD–781 Test Plan*
9	Coder-Receiver-Transmitter	KY–532A/ASQ	450	IV
10	Amplifier-Power Supply-Receiver	AM–2349/ASQ–19 or AM–2349A/ASQ–19	400	IV
11	Antenna	AS–909/ARA–48 or AS–909A/ARA–48	2500	VI
12	Control, Central	C–3076/ASQ–19	900	V
13	Control, Radio Set	C–6684/ASQ	1000	V
14	Control, Radio Set	C–6685/ASQ	5000	V
15	Intercommunication Station	LS–459/AIC	4000	VI
16	Intercommunication Station	LS–460/AIC	4000	VI
17	Indicator, Frequency Channel	ID–808/ASQ	2700	VI
18	Indicator, Frequency Channel	ID–1311/ASQ	2700	VI
19	Control, Transponder Set	C–6280(P)/APX	—	—
20	Indicator, Bearing, Distance and Heading	ID–663/U	—	—
21	Control, Radar Set	C–1159/APX–6B	—	—
22	Control, Coder Group	C–1272A/APA–89	—	—
23	Intercommunication Station	LS–459A/AIC or LS–459B/AIC	4000	VI
24	Intercommunication Station	LS–460A/AIC or LS–460B/AIC	4000	VI
25	Control, Radio Set	C–8057/ARC	—	—
26	Juliet Equipment	Juliet 28	—	—

* *Page 1*, paragraph 1.3: Add

"**1.3 Types.** The electronic Centrals covered by this specification shall include the following equipments. Each Central shall consist of the items and quantities as specified below:

* *Items Required for Each Central*

AN/ASQ–19	*AN/ASQ–19A*	*AN/ASQ–19B*	*AN/ASQ–107*
Item 1 or 2	Item 5	Item 6	Item 3 or 4
7	7	7	7
8	9	9	8
10	10	10	10

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11	11	11	11
12(2)	13(2)	13(2)	12(2)
15	14(2)	14(2)	17(2)
16	15	18(2)	20
17(2)	16	19	21
20	18(2)	20	22
21	19	23	23
22	20	24	24
		25	25
		26	26

* Item numbers correspond to items of paragraph 1.2. Quantity of each item is one unless otherwise designated by quantity in parenthesis. Items 19, 20, 21, 22, 25 and 26 are part of the electronic centrals but normally procured separately.

Page 2, paragraph 2.1 under SPECIFICATIONS MILITARY: Add

"MIL–C–38257 Control, Transponder Set C–6280 (P)/APX
MIL–R–81443 Receiver-Transmitter, Radio RT–793/ASQ; RT–793A/ASQ
MIL–C–81441 Control, Radio Set C–6684/ASQ
MIL–C–81442 Control, Radio Set C–6685/ASQ"

Page 2, paragraph 2.1, under SPECIFICATIONS MILITARY: Delete

"MIL–R–23094 Reliability Assurance for Production Acceptance of Avionic Equipment"

And under STANDARDS MILITARY: Add

"MIL–STD–781 Reliabiilty Tests, Exponential Distribution."

Page 3, paragraph 3.1.2: Delete

Page 4, paragraph 3.2.3: Delete "KY–311 (*)/ASQ–19" and substitute "KY–532/ASQ or KY–532A/ASQ".

Make this change wherever "KY–311(*)/ASQ–19" appears in this specification.

Page 4, paragraph 3.2.4.3: Add

"RT–793/ASQ
RT–546B/ASQ–19
RT–853/ASQ–19
RT–793A/ASQ
AM–2349(A)/ASQ–19"

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* *Page 4*, paragraph 3.2.4.4: Delete and substitute

"**3.2.4.4** *Reliability in Mean Time Between Failure.* The Centrals shall have a specified mean (operating) time between failures of 55 hours when tested and accepted as outlined under the requirements of 4.4.3. For individual equipments purchased separately the requirements shall be as shown in 1.2."

Page 5, paragraph 3.2.8: Delete and substitute

"**3.2.8** *Interference Control.* Unless otherwise specified in the individual equipment specification, the generation of radio interference by the equipment and the vulnerability of the equipment to radio interference shall be controlled within the limits of specification MIL–I–6181B except that an addition 69 db, 43 db, and 58 db shall be added to the limits of curves 1, 16 and 17 using a 1 KC bandwidth respectively for tests conducted in accordance with Paragraph 3.1.1 (MIL–I–6181B).

Page 5, paragraph 3.2.12: Delete "30 ± 10°C" and substitute "25 ± 5°C".

* *Pages 5 and 6*, paragraph 3.2.13: Add

"Intercommunication Station	LS–459A/AIC	Class 1
	LS–459B/AIC	Class 1
Intercommunication Station	LS–460A/AIC	Class 1
	LS–460B/AIC	Class 1
Control, Radio Set	C–6684/ASQ	Class 1
Control, Radio Set	C–6685/ASQ	Class 1
Indicator, Frequency Channel	ID–1311/ASQ	Class 1
Receiver-Transmitter, Radio	RT–546B/ASQ–19	See Section 3.2.13.3
Receiver-Transmitter, Radio	RT–793/ASQ	Class 1
Receiver-Transmitter, Radio	RT–793A/ASQ	Class 1
Receiver-Transmitter, Radio	RT–853/ASQ–19	See Section 3.2.13.3
Amplifier-Power Supply-Receiver	AM–2349A/ASQ–19	Class 2
Antenna	AS–909A/ARA–48	Class 2"

Page 7, Figure 1: Delete and substitute new Figure 1 of this amendment.

Page 8, Figure 2: Delete and substitute new Figure 2 of this amendment.

* *Page 9*, paragraph 3.3.1: Delete and substitute

"**3.3.1** *Communication Function.* The communication function performance shall meet the requirements of the Receiver-Transmitter Radio Specifications noted below for the indicated CNI systems.

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Reeciver-Transmitter Radio	*Specification Number*	*CNI System*
RT–546/ASQ–19	MIL–R–22762	AN/ASQ–19
RT–546A/ASQ–19	MIL–R–22762	AN/ASQ–19
RT–546B/ASQ–19	MIL–R–22762	AN/ASQ–107
RT–853/ASQ–19	MIL–R–22762	AN/ASQ–107
RT–793/ASQ	MIL–R–81443	AN/ASQ–19A
RT–793A/ASQ	MIL–R–81443	AN/ASQ–19B"

* *Page 9,* paragraph 3.3.1.1, line 2: Delete and substitute

"Of the RT–546A/ASQ–19 and RT–546B/ASQ–19 for the Data Link shall meet the requirements of specification MIL–R–22762"

* *Page 9,* paragraph 3.3.5: Add

"For the AN/ASQ–19 and AN/ASQ–107".

* *Page 9,* paragraph 3.3.6: Add

"for the AN/ASQ–19 and AN/ASQ–107 or the Control Radar Set C–6280(P)/APX specification MIL–C–38257 for the AN/ASQ–19A, and AN/ASQ–19B"

* *Page 9,* paragraph 3.3.7: Add

"for the AN/ASQ–19 and AN/ASQ–107."

* *Page 9,* paragraph 3.3.8: Add

"for the AN/ASQ–19 and AN/ASQ–107 or ID–1311/ASQ Specification MIL–I–22768 for the AN/ASQ–19A and AN/ASQ–19B."

* *Page 9,* paragraph 3.3.9: Delete "LS–460/AIC and substitute

"LS–460/AIC, LS–460A/AIC or LS–460B/AIC

Delete "LS–459/AIC" and substitute

"LS–459/AIC, LS–459A/AIC or LS–459B/AIC

Page 9, Add:

"**3.3.10** *Communication Control.* The remote control functions required for the operation of the communication receiver and auxiliary receiver function shall meet the requirements of Control, Radio Set C–6684/ASQ Specification MIL–C–81441 for the AN/ASQ–19A and AN/ASQ–19B.

3.3.11 *TACAN Control.* The remote control function requirement for the operation of the UHF Navigation function shall meet the requirements of Control, Radio Set C-6685/ASQ Specification MIL–C–81442 for the AN/ASQ–19A and AN/ASQ–19B.

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3.3.12 *Control, Radio Set.* The remote control function required for the operation of the Juliet equipment shall meet the requirements of Control, Radio Set C–8057/ARC Specification MIL–C–81470(AS). For the AN/ASQ–19B and AN/ASQ–107."

Page 10, paragraph 4.2.1.1, last sentence: Delete

Page 10, paragraph 4.2.3: Add

"During temperature/altitude tests outlined in Specification MIL–T–5422, steps 2, 3, 4, 6, 7, 9, 10, 11, 12, and 14; cooling air may be supplied at all times."

Page 10, paragraph 4.2.4, last sentence: Delete all after

"under the contract"

Page 12, paragraph 4.4.1.3: Delete sentence 7 and substitute

"The direction of vibration shall be either (a) vertical to the normal mounting plane for 5 minutes and lateral to the plane for 5 minutes or (b) rotary for 10 minutes."

Page 13, paragraph 4.4.3, 4.4.3.1, and 4.4.3.2: Delete and substitute

"**4.4.3** *Reliability Assurance Tests.* Reliability Assurance Tests shall be conducted using MIL–STD–781. A qualification Phase and Sampling Phase Test shall be conducted.

4.4.3.1 *Qualification Phase.* Prior to the acceptance of equipments under the contract or order, a minimum of three (3) equipments shall be tested as outlined in MIL–STD–781, under the section entitled "Qualification Phase of Production Reliability Tests." The maximum number of equipments to be used shall be those listed in Table 5 of MIL–STD–781. For the Qualification Phase, Test Level E shall be used. The Accept-Reject Criteria for Test Plan III shall be used.

4.4.3.2 *Reliability Sampling Phase Tests.* Samples of the equipment shall be tested as outlined in MIL–STD–781, under the Section entitled "Sampling Phase of Production Reliability Tests". For the Sampling Phase, Test Level E shall be used. The Accept-Reject criteria for Test Plan III shall be used to determine the length of the tests (until an accept or reject decision is reached).

4.4.3.2.1 *Lot Size for Sampling Phase.* The entire contract quantity (with the exception of those equipments used in the Qualification Phase) shall be evaluated as one lot. The test program shall start the first month after the Qualification Phase has been completed. Samples of each month's production shall be selected and placed on test in the same manner as though one month's production was a lot. The samples shall be tested until an accept or reject decision is reached. In the meantime other equipments constructed shall be shipped. The test results shall be summarized monthly for the procuring activity. The procuring activity reserves the right to stop the acceptance of equipment at any time after one or more reject decisions have been reached, pending a review of the contractor's effort to improve the equipment, the equipment parts, the equipment quality control, etc., so that the entire contract quality will show an accept decision."

* *Page 13,* paragraph 4.4.3.4(1): Delete and substitute

(1) *UHF Communications—RT–546, RT–546A, RT–546B, and RT–853.*

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Characteristics	*Specification Requirement*	*Acceptance Requirement (if lower)*
Receiver Sensitivity	Standard 5 uv/10 db Service 20 uv/10 db	20 uv/10 db
Audio Capability	Standard not less than 250 mw Service not less than 62.5 mw	Not less than 62.5 mw
RF Power Output	Standard not less than 16 w Service not less than 9 w	Not less than 9 w
Modulation Capability	Not less than 80%	Not less than 70%
Tone Modulation Capability	Not less than 70%	Not less than 70%
Side Tone Level	Standard not less than 250 mw Service not less than 75 mw"	Not less than 75 mw

Page 14, paragraph 4.4.3.4(2), after "Audio Capability": Delete and substitute

Specification Requirement	*Acceptance Requirement*
Standard not less than 250 mw Service not less than 62.5 mw	Not less than 62.5 mw

Page 14, paragraph 4.4.3.4(2): Delete

"ADF 100 cps Inverter and ADF Motor Control, and associated requirements

Page 14, paragraph 4.4.3.4(4): Delete and substitute

"(4) *TACAN–KY–312 and RT–547*

Characteristic	*Specification Requirement*	*Acceptance Requirement (if lower)*
Distance Accuracy and Sensitivity	± (0.2 mi + 0.1%) for sig. levels from −117 to −30 dbw	± (0.3 mi + 0.1%) for sig. levels from −114 to −30 dbw (20% 15 cycles, 20% 135 cycles)
Bearing Accuracy and Sensitivity	± 2° with sig. levels from −113 to −30 dbw	± 2.5° with sig. levels from −110 to −30 dbw (20% 15 cycles, 20% 135 cycles)
Identification	Standard not less than 50 mw audio with 30 db S/N ratio for sig. levels from −114 to −30 dbw	Not less than 50 mw audio with 30 db S/N ratio for sig. levels from −111 to −30 dbw

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	Service not less than 50 mw audio with 30 db S/N ratio for sig. levels from −111 to −30 dbw	
Power Output	Standard not less than 30 dbw Service not less than 27 dbw"	Not less than 27 dbw

Page 15, paragraph 4.4.3.4(5) : Delete and substitute

"(5) *IFF/Coder–KY–532/ASQ and KY–532A/ASQ*–IFF/Coder performance characteristics to be measured shall be in accordance with Specification MIL–C–22764."

* *Page 15*, paragraph 4.4.3.4: Add

"*(6) UHF Communications—RT–793 and RT–793A*

Characteristic	*Specification Requirement*	*Acceptance Requirement (if lower)*
Receiver Sensitivity	Standard 4 uv/10 db	9.9 uv/10 db
Main Receiver	Service 7 uv/10 db	
Guard Receiver	Standard 5 uv/10 db Service 10 uv/10 db	14.1 uv/10 db
Audio Capability	Standard not less than 250 mw Service not less than 79 mw	Not less than 79 mw
RF Power Output	Standard not less than 16 w Service not less than 7 w	Not less than 5 w
Modulation Capability (input to Clipping)	Not less than 80%	Not less than 70%
Tone Modulation Capability	Not less than 80%	Not less than 70%
Side Tone Level	Standard not less than 250 mw Service not less than 50 mw	Not less than 50 mw

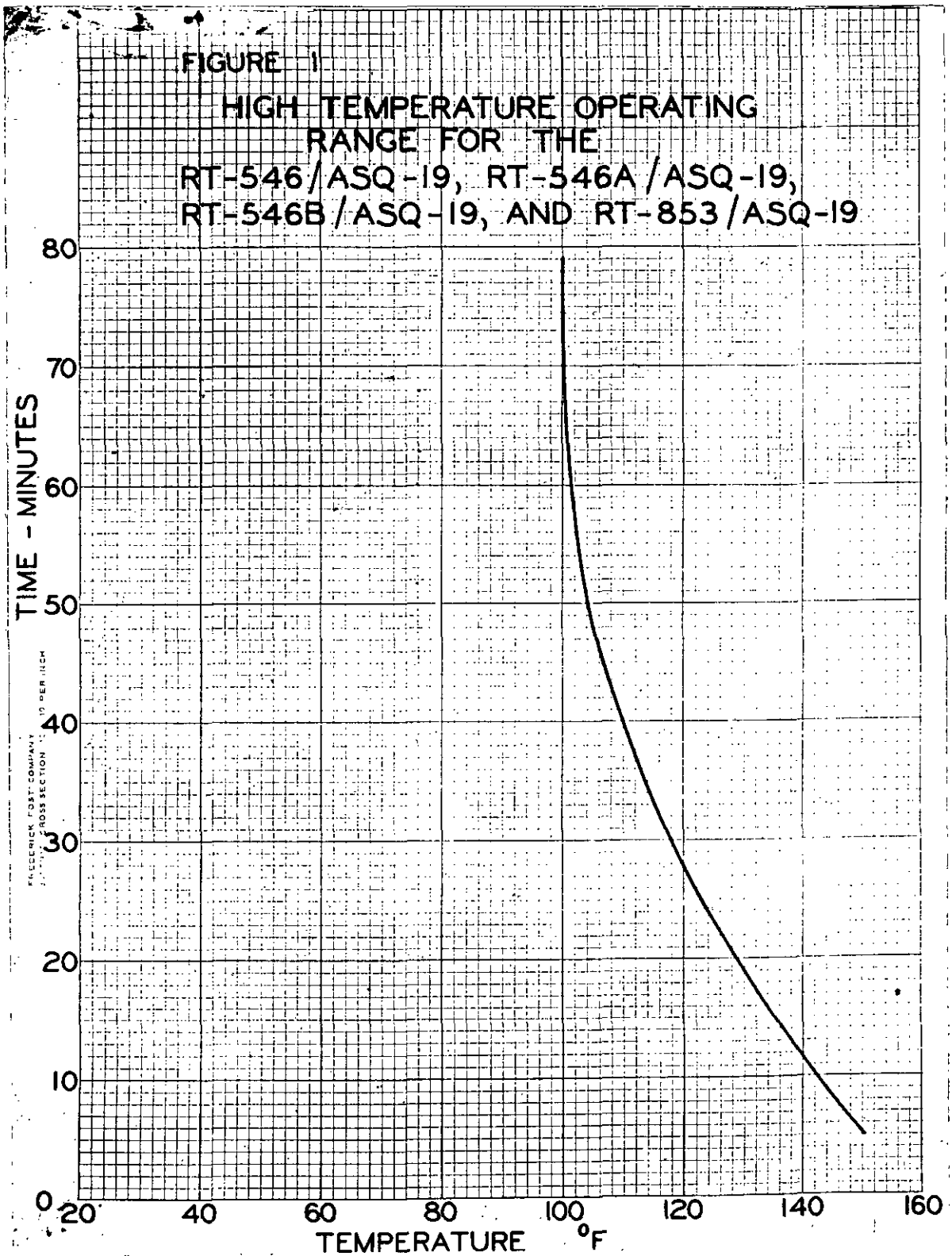
* *Page 15*, paragraph 4.4.3.6: Delete and substitute

"**4.4.3.6** *Determination of Compliance.* Compliance shall be determined in accordance with Test Plan III of MIL–STD–781."

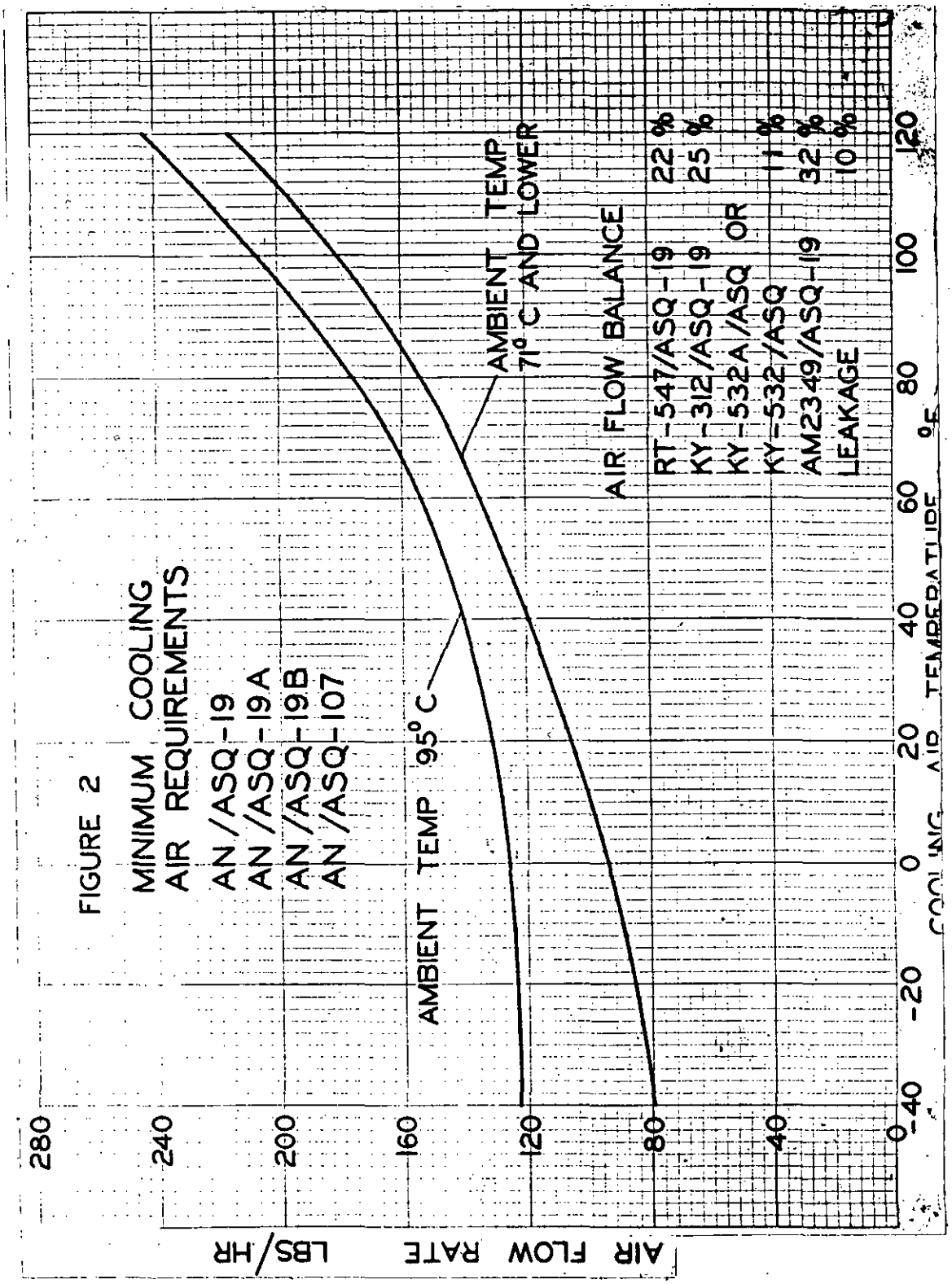
An asterisk preceding a page number denotes a change to the specification which did not appear in the previous amendment.

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FIGURE 1
HIGH TEMPERATURE OPERATING RANGE FOR THE RT-546/ASQ-19, RT-546A/ASQ-19, RT-546B/ASQ-19, AND RT-853/ASQ-19



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